

1. Research Objectives, Research Method, etc.

This research proposal will be reviewed in the Basic Section of the applicant's choice. In filling this application form, refer to the Application Procedures for Grants-in-Aid for Scientific Research-KAKENHI.
Research objectives, research method, etc. should be described within 4 pages.
A succinct summary of the research proposal should be given at the beginning.
The main text should give descriptions, in concrete and clear terms, of (1) scientific background of the proposed research, the circumstances leading to the conception of the proposed research, and the "key scientific question" comprising the core of the research plan, (2) the purpose, scientific originality, and creativity of the research project, (3) domestic and overseas trends related to the proposed research and the positioning of this research in the relevant field, (4) what will be elucidated, and to what extent and how will it be pursued during the research period, and (5) preparation status towards achievement of the purpose of the research project.

Summary of the Research Proposal

This project aims to develop the **first worldwide one-million-word error-annotated Indonesian learner corpus** by collecting, annotating, and analyzing essays written by learners of Indonesian at institutions in seven countries (Japan, USA, Italy, Czechia, Australia, Singapore, and Indonesia). Building on my ongoing JSPS Kakenhi Early-Career Scientists project (2023–2026) that developed the **Error-Annotated Indonesian Learner Corpus** or *Korpus Pemelajar Bahasa Indonesia Beranotasi Error (Koper)* from seven Japanese universities, this expanded corpus containing at least one million words will allow cross-linguistic analysis of error patterns, focusing on the influence of learners' first (L1) and additional languages (L2) on Indonesian acquisition. The dataset will be made open-access, benefiting linguists, educators, lexicographers, and AI developers. The research will provide a pioneering, empirical basis for Indonesian language pedagogy, materials development, and automated error detection systems which will benefit Indonesian language learners.

1. Scientific Background and Key Scientific Question

1.1 Scientific Background

Learner corpora—large, structured collections of authentic learner language annotated for errors—have revolutionized research and pedagogy in English and other widely taught languages such as Spanish and Japanese. However, **no comparable resource exists for Indonesian** despite its status as a major regional language in Southeast Asia (Brunei, East Timor, Indonesia, Malaysia, Singapore, and Southern Thailand). Indonesia is the world's fourth-most populous country, after China, India, and the United States. According to the 2020 census, over 97% of Indonesians are fluent in Indonesian, making it the largest language by number of speakers in Southeast Asia. Indonesian is the tenth most spoken languages worldwide in 2025 (<https://www.statista.com/statistics/266808/the-most-spoken-languages-worldwide/>). In addition, the number of **Indonesian for Foreign Speakers** or *Bahasa Indonesia untuk Penutur Asing (BIPA)* learners worldwide is increasing, with figures showing a jump from 72,746 learners in 2020 to 150,290 in 2022. It is still increasing, particularly after Indonesian was declared an official UNESCO language in 2023. This absence limits empirical research on Indonesian learner interlanguage and hampers evidence-based development of teaching materials. My current JSPS project (23K12235, 2023—2026) addresses this gap domestically by building Koper, an error-annotated corpus from essays written by over 500 Japanese learners (approx. 4,500 essays, 500,000+ words). This corpus has enabled analysis of high-frequency vocabulary and systematic affixation errors, confirming the value of annotated learner data.

1.2 Circumstances Leading to the Proposal

There are two types of learner corpus in terms of the mode of the language, i.e. written and spoken. The big goal is to build a learner corpus having both written and spoken data. To achieve this, firstly, I divide it into two stages. The first stage is to build and develop a written learner corpus and the second stage is to build and develop a spoken learner corpus. The first stage is divided into two phases. The first phase, focusing on Indonesian language learners in Japan, is my ongoing JSPS project (23K12235, April 2023—March 2026) and the second phase, focusing on Indonesian language learners worldwide, including Japan, is the current research proposal. The second stage is also divided into two phases, focusing on learners in Japan and learners worldwide. This project in general will contribute to teaching methodology, syllabus design, teaching materials, learner dictionary, development of Indonesian spelling checker, computer-assisted language learning etc., specifically to writing and grammar courses for written corpus and speaking and conversation courses for spoken corpus. The following table shows the general picture of this project.

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Error-Annotated Indonesian Learners Corpus (Koper) Project			
First Stage: Written Corpus		Second Stage: Spoken Corpus	
Phase One: Learners in Japan	Phase Two: Learners Worldwide	Phase Three: Learners in Japan	Phase Four: Learners Worldwide
Apr 2023—Mar 2026 (ongoing project)	Apr 2026—Mar 2029 (this research proposal)	Apr 2029—Mar 2032 (to be planned)	Apr 2032—Mar 2035 (to be planned)

Writing and Grammar	Speaking and Conversation
Teaching methodology, syllabus design, design of teaching materials, Indonesian Learner Dictionary, Dictionary of Common Errors in Indonesian, development of Indonesian spelling checker, computer-assisted language learning etc.	

International conferences and collaborations have connected me with BIPA instructors worldwide who are willing to contribute learner data. This opens a unique opportunity to **expand Koper into a global corpus**, enabling cross-linguistic comparison of error patterns. Given the diversity of L1 backgrounds (e.g., Japanese, English, Italian, Czech, Mandarin, Malay), such a corpus will allow us to test hypotheses about L1/L2 influence on Indonesian acquisition.

1.3 Key Scientific Question

How do the types and frequencies of grammatical, lexical, and orthographical/spelling errors in Indonesian differ according to learners' L1 and other languages learned, and what cross-linguistic patterns emerge across diverse learning environments?

2. Purpose, Scientific Originality, and Creativity

The project aims to build and analyze a large-scale, open-access, worldwide one-million-word error-annotated Indonesian learner corpus to investigate cross-linguistic influences on Indonesian acquisition. It represents the first such corpus of this scale, covering learners from multiple continents and diverse first-language backgrounds, with uniform error-tagging applied across institutions to ensure reliable cross-regional comparisons. By releasing the data openly under a CC BY license, the project seeks to accelerate research in Indonesian linguistics. Its innovative approach lies in combining corpus linguistics, second language acquisition studies, and applied pedagogy, while also designing an infrastructure compatible with CQPWeb (<https://cqpweb.lancs.ac.uk/>) and Malindo Conc (<https://malindo.aiken.jp/conc/concordance/ind/>) for public use. Moreover, the corpus has direct applications in developing AI-based automated error detection systems. Error analysis can be used to develop pedagogical resources such as textbook supplements, a learner dictionary, and a dictionary of common errors.

3. Domestic and Overseas Trends and Positioning

In recent years, quantitative research on Indonesian language teaching or BIPA has grown domestically, yet studies still rely on relatively small and localized datasets, with the Koper project standing as the only comprehensive Indonesian learner corpus to date, focused primarily on learners in Japan. Internationally, learner corpora such as International Corpus of Learner English (ICLE) and International Corpus Network of Asian Learners of English (ICNALE) for English; Corpus Escrito del Español como L2 (CEDEL2) for Spanish; and International Corpus of Japanese as a Second Language (I-JAS) for Japanese have become indispensable resources for second language acquisition (SLA) research, but no comparable resource exists for Indonesian. This project addresses that gap by establishing the first cross-linguistic Indonesian learner corpus, positioning Japan as a global leader in the field. Through this project, large-scale empirical SLA studies in Indonesian will become possible, providing a solid foundation for evidence-based Indonesian language pedagogy as well as AI-driven applications.

4. Research Plan: Scope, Methods, and Expected Outcomes

The project will shed light on comparative error profiles across different first- and second-language backgrounds, revealing how learners' linguistic origins shape their Indonesian acquisition. It will also clarify

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the role of learning environments and institutional contexts in influencing language development. Furthermore, the study will distinguish between universal error patterns common to all learners and those that are specifically tied to learners' individual L1 backgrounds. This project will create the foundational resource for Indonesian SLA research, inform pedagogical practice globally, and enable new NLP applications for Indonesian. It will also strengthen Japan's role as a hub for innovative, internationally collaborative language research. In order to achieve these goals, I plan to develop the corpus in three years.

	2026		2027		2028	
	1 st Sem.	2 nd Sem.	1 st Sem.	2 nd Sem.	1 st Sem.	2 nd Sem.
Data (+consent form) collection						
Data preprocessing & annotation						
Statistical and corpus analysis						
Academic publications						
Open corpus release						

From April 2026 to March 2029, I will collect consent forms and learners' essays from all institutions, conduct data preprocessing and annotation. During the second semester of each year, I will conduct statistical and corpus analysis. Afterwards, I will disseminate the findings through academic presentations and publications. I will release the corpus data on GitHub in the 2nd year and on online interface in the 3rd year.

Research Method

I will collaborate with Indonesian language instructors at ten institutions in seven countries, including Japan and Indonesia, to collect learner data as written in the following table. Nine institutions are universities, offering Indonesian courses as compulsory or elective subjects or as a BIPA (Indonesian for Foreign Speakers) course. The average number of learners per semester ranges from 10 in Jakarta State University to 140 in Yale University (see table below). Some learners at the universities in the USA and Australia as well as at the secondary school in Singapore are heritage learners with different writing and speaking proficiency. Thus, developing Koper will open a new research area in Indonesian language pedagogy for heritage learners.

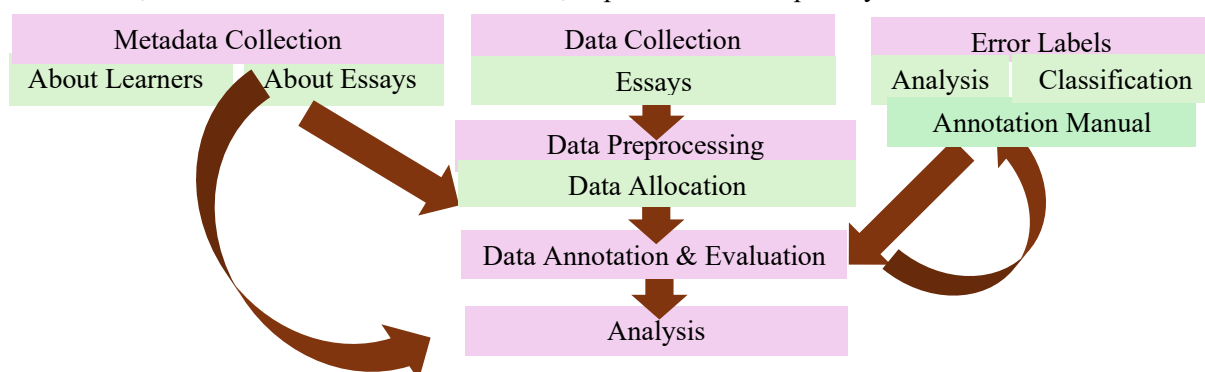
Institution	Country	Type	Course type	Learner's L1	H. L.	Number
Kanda University of Inter- national Studies (KUIS)	Japan	University	Compulsory	Japanese	none	40
Tokyo University of Foreign Studies (TUFS)						20
University of California, Los Angeles (UCLA)	Elective		Various, mainly English	some	48	
Yale University					140	
Monash University	Australia		Compulsory	Czech	none	20
Palacký University	Czechia					20
University of Naples	Italy		Elective	Italian	some	25
Victoria School	Singapore	Secondary school		Chinese, Malay, Singlish, Tamil		77
Jakarta State University	Indonesia	University	BIPA course	Various (Thai, Vietnamese, etc.)	none	10
Malang State University						15

Prior to submission, learners will be informed of the research objectives and provided with a consent form by their instructors. The form includes information such as name, age, gender, academic major, mother tongue, duration of Indonesian study, experience with other foreign languages, Indonesian proficiency tests taken, length of stay in Indonesia, and self-assessed difficulty of writing in Indonesian with justifications. Submission of the form is voluntary; essays from students who do not provide consent are excluded from the corpus. Students are also permitted to omit answers to individual questions. The background information

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gathered through the consent forms serves as metalinguistic metadata.

Error types in L2 production may be universal (e.g., typographical errors) or language-specific (e.g., verb affixation errors), depending on the influence of their L1. For this project, an error taxonomy was developed based on prior research on BIPA learner errors. Each error type was assigned a label informed by linguistic criteria and then organized into categories such as “Verb Errors,” “Word Order Errors,” and “Lexical Errors”. The taxonomy encompasses both grammatical errors and stylistic issues, including the inappropriate use of colloquial vocabulary in formal writing, overly long or convoluted sentences, and subordinate clauses without main clauses. The label set is iteratively refined during the annotation process: new labels are added when necessary and unused labels are removed. Data annotation is done by outsourcing via an IT company in Indonesia (CV IndoLeksia). Annotators report new categories to me, who evaluate their validity and inform all annotators of updates to ensure consistency. Raw essays are collected in digital formats (.pdf, .doc, .txt, or .xls, including those exported from Padlet), standardized, and converted to .txt files. Any personal information embedded in the essays (e.g., names, phone numbers, email addresses, residential addresses) is anonymized or replaced with placeholders such as <NAME>, <NUMBER>, and <EMAIL>, in accordance with the consent form, to protect learners’ privacy.



Essays are grouped and distributed evenly among annotators. Each annotator works independently on their assigned texts and remains blind to the texts annotated by others, ensuring there is no collaboration during the annotation process. Annotation is conducted using UAM CorpusTool (O’Donnell, 2008), a widely adopted software for learner corpus annotation chosen for its ease of installation, user-friendly interface, multi-layer annotation capability, metadata handling, label schema creation, corpus querying, and statistical functions. Once imported into UAM CorpusTool, essays are automatically segmented into sentences (one sentence per line), enabling annotators to concentrate on local errors while still viewing the full essay for contextual interpretation. Annotators are instructed to label errors as precisely as possible. Labels could be assigned to single words, multiword expressions, or entire sentences. Overlapping errors can be annotated within the same sentence, and multiple error labels can be assigned to a single word if necessary. Missing words or punctuation can also be annotated by selecting the surrounding context where the element should have occurred. Upon completion of each annotation task, annotators save their files and export the results. I compare the results across annotators. This comparison enables the evaluation of annotation consistency and the calculation of inter-annotator agreement. Afterwards, analysis such as frequency analysis of error types per L1 group, statistical modeling of L1/L2 influence (logistic regression, correspondence analysis), and comparative analysis with Japanese learner data will be conducted.

5. Preparation Status

The project builds on a solid foundation of preparation in terms of infrastructure, collaboration, and expertise. Annotation protocols, tagsets, and workflows have already been established through the current Koper project, providing a tested technical framework for large-scale learner corpus development. Internationally, agreements in principle have been secured with institutions across seven countries, and active contacts with BIPA instructors have been established to facilitate data collection. The research team brings extensive experience in constructing and annotating large-scale learner corpora as well as conducting SLA-focused analyses. Furthermore, preliminary results from published studies on high-frequency vocabulary and affixation errors in Japanese learner data underscore both the feasibility of the approach and its scholarly impact, ensuring that the proposed project is well-positioned for successful implementation.

2. Applicant's Ability to Conduct the Research and the Research Environment

Descriptions of (1) applicant's hitherto research activities (including main research achievements), and (2) research environments including research facilities and equipment, research materials, etc. relevant to the conduct of the proposed research should be given within 2 pages to show the feasibility of the research plan by the applicant (Principal Investigator).

If the applicant has conducted any international efforts related to his/her research plan (such as his/her records of joint international research and research history in overseas institutions), they should be included as necessary in "(1) applicant's hitherto research activities". Also, if the applicant has taken leave of absence from research activity for some period, he/she may choose to write about it therein.

(1) Applicant's Hitherto Research Activities and Achievements

Research Experience and Domain Expertise

I got my Ph.D. degree from NTU, Singapore in 2018 and currently serve as Associate Professor in Indonesian Major at the Department of Asian Languages, Kanda University of International Studies (2025–present), following a four-year appointment as full-time lecturer there (2021–2025). Before that (2019–2021) I worked as Assistant Professor in Indonesian Studies at the Department of Asian Studies, Faculty of Arts, Palacký University Olomouc in Czechia. My expertise spans Indonesian grammar, computational linguistics, corpus linguistics, lexicography, and BIPA (Indonesian for Foreign Speakers) pedagogy. I am the Principal Investigator of Kakenhi (Early Career) project 23K12235 (2023–2026): *Building an Error-Annotated Corpus of Learner Indonesian and Developing an Automated Writing Support for Japanese Students Using Deep Linguistic Indonesian Parsers*. At the same time, I am one of the co-investigators of Kakenhi (General) project 23H00639 (2023–2027): *Building a Corpus of Linguistic Acceptability for Malay/Indonesian*.

(A) Indonesian Grammar

I have done many research projects related to Indonesian language and linguistics, as follows.

Publication 1: Moeljadi, David (2024) Onomatopoeia in Indonesian. In L. Körtvélyessy & P. Štekauer (Eds.), *Onomatopoeia in the World's Languages: A Comparative Handbook* (pp. 837–848). Berlin, Boston: De Gruyter Mouton. (<https://doi.org/10.1515/9783111053226-070>)

Summary: I examined the characteristics and functions of onomatopoeia in Indonesian. The study provides an overview of the phonological, morphological, and semantic patterns of Indonesian onomatopoeic forms, as well as their role in everyday communication and literary expression. By situating Indonesian within a cross-linguistic perspective, this work highlights both language-specific features and broader typological tendencies. This publication demonstrates my expertise in descriptive and comparative linguistics, particularly in the areas of sound symbolism and word formation.

Publication 2: Moeljadi, David and Viola Ow (2018) Serial Verb Constructions in Indonesian: An HPSG Analysis and Its Computational Implementation. *Journal of the Southeast Asian Linguistics Society (JSEALS) Special Publication*, (2), 90–101.

Summary: I investigated serial verb constructions (SVCs) in Indonesian through the framework of Head-driven Phrase Structure Grammar (HPSG). The research not only provides a formal syntactic analysis of SVCs, but also demonstrates their computational implementation, bridging theoretical linguistics and natural language processing. By modeling Indonesian SVCs in HPSG, we showed how linguistic theory can contribute to computational grammar development.

(B) Learner Corpus Development

I led the development of the error-annotated Indonesian corpus and analyzed the data.

Publication 1: Moeljadi, David (2023) Penyusunan KOPER: Korpus Pemelajar Bahasa Indonesia Beranotasi Error [Building KOPER: an Error-Annotated Indonesian Learner Corpus]. In Adi Budiwiyanto et al. (Eds.), *Prosiding Kongres Bahasa Indonesia XII*, pp. 429–444. Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi.

Summary: I built KOPER (Korpus Pemelajar Bahasa Indonesia Beranotasi Error), the first error-annotated learner corpus of Indonesian. The study outlines the process of compiling and annotating essays written by learners of Indonesian at various universities in Japan, focusing on the systematic tagging of linguistic errors. Beyond its value as a linguistic resource, KOPER serves as an empirical foundation for research on second language acquisition and for the development of pedagogical materials in BIPA (Indonesian for Foreign Speakers).

Publication 2: Moeljadi, David (2025) Kata Berfrekuensi Tinggi dalam Korpus Pemelajar Bahasa Indonesia Beranotasi Error [エラータグ付きインドネシア語学習者コーパスにおける高頻度語]. In Yasuda

[2. Applicant's Ability to Conduct the Research and the Research Environment (continued from the previous page)]

Kazuhiko (ed.) *Bahasa dan Budaya: Jurnal Himpunan Peneliti Indonesia Seluruh Jepang* No. 30 [インドネシア言語と文化第30号]. Himpunan Peneliti Indonesia Seluruh Jepang [日本インドネシア学会].

Summary: I used the Korpus Pemelajar Bahasa Indonesia Beranotasi Eror (KOPER) to analyze vocabulary usage in learner essays collected from seven universities in Japan between April 2023 and March 2024. The findings show that high-frequency words in learner writing overlap with those in general Indonesian corpora and BIPA conversational corpus, but also reveal distinctive features such as the predominance of formal written style and frequent use of the first-person singular pronoun *saya*, reflecting common essay topics like self-introduction and personal opinions. The analysis further demonstrates that beginner learners (level one) rely mainly on basic nouns and verbs, while intermediate and advanced learners (level two and above) employ a wider range of nouns, verbs, adjectives, and adverbs, as well as more frequent use of conjunctions, prepositions, the relative pronoun *yang*, and the copula *adalah*. These results highlight how vocabulary use is shaped not only by learners' proficiency levels but also by essay themes, and they provide valuable implications for the development of teaching materials, especially in determining the list of words that must be taught according to the level.

(C) Computational Tools Development

I led or contributed to the development of several Indonesian linguistic resources, as follows.

Publication 1: Kamajaya, Ian and David Moeljadi (2025) IndoMorph: a Morphology Engine for Indonesian. In *Proceedings of the Second Workshop in South East Asian Language Processing*, pages 72–81, Online. Association for Computational Linguistics.

Summary: The study presents the design and implementation of a rule-based system capable of analyzing and generating Indonesian word forms, covering a wide range of morphological processes such as affixation and reduplication. By integrating linguistic insights with computational methods, IndoMorph demonstrates how formal linguistic knowledge can be operationalized in natural language processing. This work highlights my expertise in morphology and resource development for Indonesian, and contributes to advancing tools that support both linguistic research and practical applications.

Publication 2: Nomoto, Hiroki, David Moeljadi and Farhan binti Abdul Razak (2024) Masalah teknologi dan isu sosial berkaitan penggunaan ChatGPT dalam bahasa Melayu [Technological and social issues related to the use of ChatGPT in Malay]. *RENTAS: Jurnal Bahasa, Sastra dan Budaya* 3: 1-22.

Summary: The paper discusses the challenges of applying large language models to Malay, including limitations in training data, linguistic accuracy, and adaptability to local usage contexts. It also addresses broader societal concerns such as ethical implications, language preservation, and the potential impact of generative AI on education and communication. I commit to ensuring that technological innovations are critically examined in relation to less-resourced languages like Malay and Indonesian.

Capability Summary: I bring a strong track record of corpus development, computational tools, and Indonesian linguistics. My leadership in building the KOPER and release of resources under open license (e.g., GitHub sources) demonstrates both capacity and commitment to impactful research.

(2) Research Environment, Facilities, and Materials

Kanda University of International Studies provides strong institutional support and opportunities for both domestic and international dissemination of research. The project builds on an established technical pipeline for corpus preprocessing, error annotation with UAM CorpusTool, and database management. Existing resources include the current Koper corpus—comprising approximately 3,000 essays (~350,000 words) from over 500 learners in Japan, with ongoing annotation and established metadata protocols. A strong collaborative network is already in place with BIPA instructors at institutions in the US, Japan, Italy, Czechia, Australia, Singapore, and Indonesia, securing preliminary data access and institutional backing. Human resources are also well supported, with trained annotators from an outsourcing IT company in Indonesia (CV IndoLeksia), teaching assistants, and graduate students available, with existing guidelines and workflows. The project has a proven record of open science, with prior releases of Koper under a CC-BY-SA-4.0 license on GitHub (<https://github.com/davidmoeljadi/Koper>), and a clear plan for dissemination via public platforms like CQPWeb and Malindo Conc. Taken together, the combination of institutional support, computational infrastructure, annotated data, methodological expertise, and international collaboration ensures an exceptionally strong research environment, making the successful execution of the proposed global learner corpus project within the grant period highly feasible.

3. Issues Relevant to the Protection of Human Right and Compliance with Laws and Regulations

(cf. Application Procedures for Grants-in-Aid for Scientific Research)

If the proposed research involves such issues that require obtaining the consent and/or cooperation of third parties, consideration in handling of personal information, or actions related bioethics and/or biosafety, including the laws, regulations and the guidelines in the country/region(s) where the joint international research is to be conducted, describe the measures and actions planned to be taken in responding to these issues within 1 page.

This provision applies to research activities that would require approval by an internal or external ethical jury, such as research involving handling of personal information from questionnaire surveys, interviews and/or behavior surveys, including personal histories and images, handling of donated specimens, human genome analysis, recombinant DNA, and experimentation with animals.

If the activities of the proposed research do not fall under such categories, enter "N/A (not applicable)".

I will collect and use data from the learners or research collaborators, i.e. students' writing assignments, subject to their consent. I will follow the rules and regulations of each institution regarding handling personal information etc. I will conduct my research after I get permission from each institution.

Students' writing assignments involving personal information will be handled based on the approval of the students. In conducting the data gathering, the following measures will be taken after fully recognizing whether the data is essential for carrying out the research plan and whether the collaborators (students) will not be identified by publishing the research results.

(1) Prior approval by the Research Misconduct Prevention Committee

In accordance with the "Code of Ethics and Ethical Review for Research Involving Human Subjects" of Kanda University of International Studies, to which the applicant belongs, this research will be conducted after the "Application for Research Ethics Review" has been submitted to the Research Misconduct Prevention Committee and approval has been obtained from that committee.

(2) Building a relationship of trust and accountability with research collaborators (students)

The applicant will clarify the research purpose and research plan, and then carefully explain in writing or orally that the data is truly necessary to carry out the research plan, until the applicant obtains the understanding of the collaborators (students). The collaborators (students) will be informed of the need to be careful about the wording of the assignments. In addition, it will be clearly communicated that giving the assignment data is voluntary and that any parts in the assignment that they do not wish to share need not be provided. After the data has been collected, the progress of the research will be communicated to the collaborators (students) as necessary, and the collaborators (students) should be informed in advance of any research presentations, so that a relationship of trust can be built with the collaborators (students) and accountability to them can be fulfilled.

(3) Maintaining the confidentiality of research collaborators (students)

I will use passwords for storage media and strictly manage personal information to ensure that the collaborators (students) do not suffer any disadvantages as a result of their cooperation in the research project. I also guarantee that the data will not be used for any purpose other than the relevant research.